Multivariate Analysis of the Physicochemical Characteristics of Portuguese Vinho Verde Wines

Introduction

Wine quality is easily the most important aspect in the wine industry, in no part due to its association with the high class, and therefore its chemical composition are the subjects of significant interest in both the agricultural sector and the data science community. The physicochemical characteristics of wine, such as acidity, sulfur dioxide levels, and alcohol content, dictate its shelf life, physical body, and sensory profile. The purpose of this study is to conduct a comprehensive multivariate analysis to understand the fundamental chemical differences between red and white wines, reduce the dimensionality of these chemical measurements, and uncover the latent factors that make up wine composition.

Dataset

This analysis utilizes the Wine Quality Dataset sourced from the UCI Machine Learning Repository, originally collected by Cortez et al. (2009). The raw dataset contains a total of 6,497 observations, consisting of 1,599 red wine samples and 4,898 white wine samples of the Portuguese "Vinho Verde" variant. The data includes 11 continuous physicochemical variables and two categorical variables: a sensory quality score and wine color. No missing values are present in the dataset, but 1,177 duplicate observations were identified and removed prior to analysis to ensure data integrity, resulting in a final count of 5,320 observations.

Research Questions and Hypotheses

This study addresses three primary research questions:

Research Question 1 (MANOVA): Do red and white wines differ significantly in their overall physicochemical characteristics when several characteristics are considered simultaneously?

- H_0: The mean vector of the selected physicochemical variables is the same for red and white wines.
- H_1: The mean vectors differ between red and white wines.

Research Question 2 (PCA): Can the 11 physicochemical measurements of wine be summarized using a smaller number of principal components while retaining a substantial proportion of the total variability?

Research Question 3 (EFA): What underlying latent dimensions or factors explain the relationships among the physicochemical characteristics of wine?

Methodology

Data preparation and statistical modeling were conducted using Python, using the pandas, numpy, scipy, statsmodels, scikit-learn, and factor_analyzer libraries. The red and white wine datasets were combined, and duplicate observations were removed.

Exploratory Data Analysis (EDA) was conducted using descriptive statistics, histograms, and boxplots to assess distributions, along with a correlation matrix to examine relationships. For the Multivariate Analysis of Variance (MANOVA), four variables (volatile_acidity, chlorides, fixed_acidity, sulphates) were selected because examining the boxplots revealed that they possessed the most pronounced median differences between the wine colors. Pillai's Trace was chosen as the most reliable metric for the MANOVA due to its robustness against multivariate non-normality and heteroscedasticity.

For Principal Component Analysis (PCA) and Exploratory Factor Analysis (EFA), all 11 physicochemical variables were standardized (Z-score normalization) due to the differing scales the variables were measured by, this prevents variables with larger measurement scales from dominating the variance. PCA components were retained based on the eigenvalue > 1 criterion and visual inspection of a scree plot. EFA factorability was assessed using Bartlett's Test of Sphericity and the Kaiser-Meyer-Olkin (KMO) measure. EFA was extracted using Principal Axis

Factoring to specifically isolate shared variance. A Promax (oblique) rotation was applied because the exploratory correlation matrix had shown that the certain variables (such as sugar, density, and dioxide) were naturally correlated.

Results

Exploratory Data Analysis

Descriptive statistics revealed that total_sulfur_dioxide (mean = 114.11, std = 56.77) and free_sulfur_dioxide (mean = 30.04, std = 17.81) exhibit the highest variability among all measurements. Correlation analysis identified strong positive relationships between residual_sugar and density (r = 0.52), and strong negative relationships between density and alcohol (r = -0.67). Boxplots demonstrated that white wines generally maintain higher concentrations of total sulfur dioxide, whereas red wines display notably higher volatile acidity, chlorides, and sulphates. Otherwise, no other observations or variables were deemed of interest.

MANOVA

Assumption checks indicated that the data deviated from strict multivariate normality (Shapiro-Wilk $p < 0.001$ for all variables) and homogeneity of variances (Levene's Test $p < 0.001$ for all variables). Additionally, Mahalanobis distances flagged 145 potential multivariate outliers ($p < 0.001$). These violations are common in large, natural datasets, thus justifying the use of the robust Pillai's Trace statistic.

The MANOVA yielded a statistically significant result (Pillai's Trace = 0.6308, $F(4, 5315) = 2270.1359, p < 0.001$). This represents a strong effect size, with wine color explaining approximately 63.1% of the variation across the four chemical characteristics chosen. Follow-up univariate ANOVAs confirmed that volatile_acidity, chlorides, fixed_acidity, and sulphates all individually contribute significantly to this distinction ($p < 0.001$ for all). In the context of wine chemistry, this confirms that the distinction between red and white wines goes far beyond just its color; they operate on fundamentally different chemical compositions especially in regards to their core acidic structure (fixed and volatile acidity), mineral extraction (chlorides), and the concentration of sulfur-based compounds (sulphates).

PCA

Three components yielded eigenvalues greater than 1 (PC1 = 2.989, PC2 = 2.476, PC3 = 1.588). PC1 explained 27.17% of the variance, PC2 explained 22.50%, and PC3 explained 14.43%, cumulatively retaining 64.10% of the total variability.

- **Component 1 (Preservative Profile):** Captured high positive loadings for total_sulfur_dioxide (0.476) and free_sulfur_dioxide (0.426), representing the chemical preservatives utilized for shelf stability.

- **Component 2 (Fermentation Body):** Captured high positive loadings for density (0.565) and residual_sugar (0.368), paired with a heavy negative loading for alcohol (-0.465), representing the physical fermentation trade-off.

- **Component 3 (Acidic Profile):** Captured high positive loadings for citric_acid (0.577) and fixed_acidity (0.424), and a negative loading for ph (-0.465), representing the acidity and sourness of the wine.

EFA

Bartlett's Test of Sphericity was highly significant (Chi-Square = 29789.89, $p < 0.001$), indicating sufficient correlation among the variables to proceed. However, the Kaiser-Meyer-Olkin (KMO) Measure was calculated at 0.4013. Because this is below the generally accepted threshold of 0.50, it indicates that the variables share a high degree of unique variance.

Despite the low KMO, EFA was conducted for exploratory purposes. The Kaiser criterion (eigenvalues > 1) and the scree plot suggested retaining three factors (eigenvalues: 2.988, 2.475, and 1.587). Three factors were extracted using Principal Axis Factoring and rotated via Promax (oblique):

- **Factor 1 (Salts & Spoilage vs. Preservatives):** Heavily loaded by chlorides (0.656) and sulphates (0.615), alongside volatile_acidity (0.535), with a strong negative cross-loading for total_sulfur_dioxide (-0.691).

- **Factor 2 (Fermentation Body & Acidity):** Showcases strong positive loadings for density (0.843), fixed_acidity (0.762), and residual_sugar (0.704), and a heavy negative loading for alcohol (-0.816).

- **Factor 3 (The Sourness):** Driven by a strong positive loading for citric_acid (0.859) and a strong negative loading for ph (-0.726).

Integrated Discussion

The findings across all three multivariate techniques yield a cohesive understanding of Vinho Verde wines, addressing each of the core analytical goals:

- **MANOVA Distinctions:** When analyzed simultaneously, red and white wines are fundamentally distinguished by their differing balances of volatile acidity, fixed acidity, chlorides, and sulphates.
- **PCA Major Dimensions:** The major dimensions of total variation among the 11 measurements can be summarized into three main components: a Preservative Profile (sulfur dioxides), a Fermentation Body (density, sugar, alcohol), and an Acidic Profile (citric acid, fixed acidity, pH).
- **EFA Latent Factors:** EFA revealed that the shared variance among the variables is explained by three latent factors: a "Salts & Spoilage vs. Preservatives" factor, a "Fermentation Body & Acidity" factor, and a "Sourness" factor.
- **Comparison of PCA and EFA:** Both techniques successfully reduced the data into three primary structures. The key similarity is that both PCA and EFA consistently grouped density, residual_sugar, and alcohol together as a core factor. The key difference lies in their methods: PCA created perfectly independent components to explain total variance, while EFA utilized an oblique (Promax) rotation to explore shared variance. This EFA approach revealed that these chemical traits actually correlate in real life (e.g., total_sulfur_dioxide loaded strongly on Factor 1, but in a negative direction compared to chlorides and volatile_acidity).
- **Substantive Interpretation:** Ultimately, these findings tell us that the physicochemical characteristics of Vinho Verde wines are not random. Instead, wine chemistry can be sectioned into three different factors, its fermentation body, its preservation methods, and its sourness/acidity.

Limitations: Two significant limitations must be noted regarding this study. First, the dataset features highly imbalanced group sizes (1,599 red vs. 4,898 white), which can skew pooled covariance matrices. Second, the low Kaiser-Meyer-Olkin (KMO) Measure (0.4013) indicates poor overall factorability. This means that not all 11 variables group well together. Trying to force every variable into one EFA model created somewhat fragmented factors, suggesting that future studies could achieve clearer results by simply dropping the least related variables before running the extraction.

Conclusion

This study successfully applied multivariate techniques to dissect the Wine Quality Dataset. Red and white wines exhibit clear, statistically significant differences in their baseline chemical profiles. Furthermore, the complexity of 11 distinct physicochemical measurements can be reliably reduced to three highly interpretable components or latent factors, demonstrating that wine composition is fundamentally governed by dimensions of preservation, body, and acidity.

References

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems*, 47(4), 547-553.

UCI Machine Learning Repository. (n.d.). *Wine Quality Dataset*. Retrieved from https://archive.ics.uci.edu/dataset/186/wine+quality